September 9, 2026

Dear Editor-in-Chief,

We are pleased to submit our manuscript, "Physics-Exact Credit Assignment for Multi-Agent Flutter Suppression," for consideration as a Research Article in the Journal of Sound and Vibration.

This manuscript was previously submitted to two AIAA journals, each of which returned it as outside their scope, for opposite reasons. The Journal of Guidance, Control, and Dynamics withdrew it as insufficiently narrow to guidance, navigation, and flight control in the sense that journal covers. AIAA Journal subsequently returned it as outside its scope of fundamentals/methods contributions because of its applied nature, and suggested the authors consider a relevant journal from the manuscript's own references. We agree with that assessment: this is an applied study of active flutter suppression on a specific aeroelastic testbed, in the same genre as prior work we cite that appeared in this journal (Mu et al., "Machine learning-based active flutter suppression for a flexible flying-wing aircraft," JSV 529, 2022), and we believe the Journal of Sound and Vibration, rather than a fundamentals-only or guidance-and-control-only venue, is the right home for it.

The paper studies active flutter suppression on a wing with several independently actuated trailing-edge surfaces, each commanded by its own reinforcement-learning agent from local sensing alone – the setting a physically distributed avionics architecture actually presents, as opposed to the single-agent, single-effector, globally-observed setting the existing flutter-suppression literature uses throughout. Posing the problem this way exposes a credit-assignment question with a physical answer: for this class of energy-conserving structural plant, we show that the per-agent credit multi-agent reinforcement learning normally has to estimate is instead available in closed form directly from the equations of motion, and we validate it against a naive undecomposed reward and an adapted value-decomposition baseline (Holm-Bonferroni corrected across the full comparison family), against a six-rung classical baseline ladder (including a fully decentralised LQG synthesised from the same sensing the learned agents have), and against an unsteady vortex-lattice solver for the underlying plant. A second, more general finding follows from the same study: on this problem, the initial policy-exploration scale dominated every algorithmic choice we tried, by a factor of two, and is computable in advance from a classical design – a diagnostic we believe generalises to other physical control tasks with a narrow useful actuation band. We report two negative results (learned communication did not help; a shaping-based refinement did not survive correction for multiple comparisons once compared against the full family of alternatives) at the same statistical standard as the positive ones.

We believe this manuscript is a strong fit for the Journal of Sound and Vibration: it is a validated structural-dynamics and vibration-control study – a Goland wing modelled from its equations of motion, cross-checked against an unsteady vortex-lattice solver, with active suppression compared against a six-rung classical control ladder – using multi-agent reinforcement learning as the control method under test rather than as the paper's primary contribution in its own right. It also engages directly with the certification gap between a learned policy and a classical design with a known stability margin, which we discuss explicitly in the paper.

Other than the two AIAA submissions disclosed above, this manuscript, in whole or in substantial part, has not previously been submitted elsewhere. I will provide any further disclosure the editorial system requests. I declare no conflicts of interest, and no funding was received for this work. Data and code needed to reproduce every result are cited in the manuscript's Data and Code Availability section.

Thank you for your consideration. I look forward to your response.

Sincerely,

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